

Practical Guide: Active Q-Control on Zurich Instruments UHF

Based on notes from R. Stomp see <https://www.zhinst.com/europe/en/blogs/resonance-engineering-quality-factor-q-control-method>

Application: QCM / Resonator Tracking with Active Damping

Instrument: Zurich Instruments UHFLI (UHF-Lock-in)

Topology: 4-Demodulator "Mnemonic" Setup (1=PLL, 2=Measurement, 3=Q, 4=AGC)

Part 1: The Experimental Workflow

This section outlines the logical sequence to initialize the system, calibrate the active damping, and lock the resonator for measurement.

Step 1: Initial Sweep & Characterization

Before engaging any control loops, we must find the natural state of the QCM.

Run a frequency sweep across the expected resonance (e.g., around 6 MHz).

Calculate the native resonance frequency (f_0), bandwidth (BW), and native Q-factor from the amplitude peak.

Locate the frequency where the phase slope ($d\Theta/df$) is steepest.

Record the exact phase angle at this steepest point. **This is the resonance phase setpoint.**

(Lorentz fit and Pratt circle should give similar values; use unwrap in sweep ?)

Check the slope direction: If it goes down through resonance, the slope is Normal. If it goes up (due to parasitic capacitance), the slope is Inverted.

Step 2: Engage Frequency Tracking (PID 1)

The Phase Locked Loop (PLL) must be engaged first and remain active continuously.

Use the PID Advisor to calculate baseline Proportional (P) and Integral (I) gains for a target bandwidth of 10 to 50 Hz.

Apply the polarity rule: If your phase slope is Normal, use Negative P and I gains. If Inverted, use Positive P and I gains. **D-Gain is always 0.**

Apply your measured phase setpoint from Step 1 to PID 1.

Enable PID 1 (in PLL). The lock-in will now continuously track the resonance frequency.

Step 3: Q-Calibration & Scanning (PID 3)

With the frequency locked, we must calibrate the relationship between the Q-Control P-gain (K_p) and the physical damping (Γ). By stepping the drive amplitude to half instead of zero, the resonator undergoes a transient decay to a new steady state. Because the signal never drops into the noise floor, the PLL remains actively locked throughout the measurement.

Ensure PID 3 P-gain starts at 0.

Define a safe K_p scan array (e.g., 0.0 to 10.0).

For each K_p value in the array:

Apply K_p to PID 3, **enable PID 3** (`pids/2/enable -> 1`), and wait 500 ms for the system to settle.

Subscribe to the Demod 2 data stream via software and begin polling (or use DAQ or scope?).

Step the drive signal amplitude down to 50% of its initial value (do not disable the output).

Measure ~30 - 50 ms for the amplitude to decay to the new half-amplitude steady state.

Restore the drive signal amplitude to its full original value and stop polling.

Disable PID 3 (`pids/2/enable -> 0`) before iterating to the next gain value.

Wait approximately 3 to 5 times the time constant (τ) for the physical amplitude to stabilize; 10 msec is enough.

Verify that the PLL remained securely locked during the transient step.

Safety Check: If the amplitude increases instead of decaying, we have crossed the lasing threshold. Abort the scan and force PID 3 to 0. Note : in the v3 of the program, an estimate of the lasing limit is made before increasing too much the K_p .

Fit the valid decay curve to extract the time constant (τ) and calculate the damping rate (Γ).

Verify that increasing K_p reduces the transient time constant before proceeding.

Step 4: Engage Steady State

Using the linear fit of the calibration data, we calculate the K_p required for the target Q (e.g., 10k).

Apply this final K_p to PID 3 and **enable PID 3 permanently** (`pids/2/enable -> 1`). The resonator should now actively be damped.

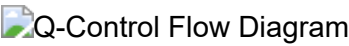
Scale the PLL: Because lowering the Q widens the bandwidth and alters the phase slope, scale the

PID 1 gains to maintain stability: $P_{new} = P_{native} \times \frac{Q_{native}}{Q_{target}}$.

Engage AGC: Enable PID 4 (`pids/3/enable -> 1`) to adjust the drive voltage and keep the amplitude stable at the target setpoint.

Begin the mass-loading experiment, logging the Oscillator 1 frequency continuously.

Q-Control Signal Flow Diagram



Part 2: Hardware & System Topology

A. Hardware Wiring & Output Routing

Input Signal (`Signal Input 1`): Resonator response. Feeds all four Demods.

Drive Output (`Signal Output 1`): Main excitation, controlled by PID 4 (AGC). Set Phase to 0.0 deg. Routed to Oscillator 1.

Q-Feedback (`Signal Output 2`): Feedback force, controlled by PID 3 (Q-Control). Set Phase to 90.0 deg. **Must be explicitly routed to Oscillator 1.**

***Critical Hardware Step:** Physically sum `Signal Output 1` and `Signal Output 2` using a BNC T-piece before connecting to the resonator.*

B. Demodulator Settings (Example for 6 MHz, Q=50k)

All Demods must be referenced to `Oscillator 1`. To optimize USB bandwidth, enable data streaming ONLY for Demod 2 during the measurement phase.

Demodulator	Role	Time Constant (TC)	Filter Order
Demod 1	PLL Source	~1 ms	4 (Standard)
Demod 2	Measurement	10 μ s	1 (Crucial to prevent filter delay)
Demod 3	Q-Control	~30 μ s	4 (Fast)
Demod 4	AGC Source	~100 ms	4 (Slow)

C. Controller Configuration Reference

Configure the PID modules in your API. Note that PID 2 is skipped.

Parameter	PID 1	PID 3 (Q-Control)	PID 4 (AGC)
Function	Frequency Tracking	Active Damping	Amplitude Stability
Module Mode	PLL	PID	PID
Input Source	Demod 1 Phase	Demod 3 R (Amp)	Demod 4 R (Amp)
Setpoint	Measured Phase at Resonance	0.0 V	Target Amp (the measured R value at demod 4)
Output Channel	Oscillator 1 Freq	Signal Output 2 Amp	Signal Output 1 Amp
Output Phase	N/A	90.0 deg (Crucial)	0.0 deg
P-Gain (Kp)	Adaptive & Sign-Dependent	Variable (Scan this)	Strictly Positive
I-Gain (Ki)	Sign-Dependent	Zero	Strictly Positive (Slow)
D-Gain (Kd)	Zero (Do not use)	Zero	Zero

Critical Note on PID Polarity: If parasitic capacitance inverts the phase slope, we must invert the P and I gains for **PID 1 only**. PID3 and PID4 normally use positive polarity, but verify the Q-control sign by checking that increasing K_p increases damping (shorter τ).

Part 3: Mathematical & Software Reference

A. The Step-Response Fit & Data Slicing

To extract the Q-factor, fit the transient amplitude envelope data from Demod 2 to:

$$A(t) = A_0 \cdot e^{-t/\tau} + C$$

Here, C represents the new 50% steady-state amplitude baseline rather than the noise floor, ensuring the PLL signal remains robust.

Data Slicing via Sliding-Window Knee Detection:

Standard curve-fitting functions fail or become unstable if fed the pre-trigger hardware delay flatline. To isolate the pure decay curve robustly I tested several methods like derivative, fitting t_0 ,... and it appears the Knee Detector works best:

- Define Windows:** Choose a sliding window size (N) of roughly 20 to 30 points (or approximating a few time delays of Demod 2 filter).
- Calculate Local Slopes:** Slide two adjacent windows of size N across the amplitude array. Fit a simple linear regression to each window to extract their local slopes (m1 and m2).

Find the Knee (t0): Calculate the difference between the adjacent slopes ($\Delta m = m_2 - m_1$).

Search the array for the index where this difference is maximized. This pinpoints the corner where the flatline drops into the steep exponential transient.

Slice & Zero: Truncate the time and amplitude arrays starting from this index (or shift 5 to 10 points to the right to completely clear the filter's rounded shoulder). Subtract the first time value from the entire sliced time array so the pure transient fit starts at $t=0$.

Calculate Damping: $\Gamma = \frac{1}{\tau} = \frac{\pi \cdot f_0}{Q}$

B. Calibration and Dynamic Safety Logic

The relationship between PID 3 P-gain (K_p) and system damping (Γ) is linear:

$$\Gamma(K_p) = \Gamma_{native} - \alpha \cdot K_p$$

Where Γ_{native} (the intercept) and α (the slope) are obtained by fitting the extracted damping rates against the scanned K_p values.

Dynamic Lasing Threshold Estimation:

If K_p is set too high, the total damping becomes negative, causing the amplitude to grow exponentially (Lasing). To avoid blindly stepping into this dangerous regime during an automated scan:

Measure the natural damping Γ_{native} at $K_p = 0$.

Perform 2 to 3 initial partial ring-downs at small, strictly safe K_p increments (e.g., 0.5, 1.0).

Calculate a preliminary linear fit on these initial points to estimate the coupling slope (α).

Dynamically predict the maximum safe gain: $K_{pmax} = \Gamma_{native} / \alpha$.

Programmatically limit the remainder of your K_p scan array to stay safely below this threshold (e.g., cap the maximum scan value at 80% or 90% of K_{pmax}).

Reaching the Target Q:

Once the fully bounded scan is complete and your final α is accurately fitted, calculate the exact K_p required for your target Q:

$$K_{p,target} = [\Gamma_{native} - (\pi \cdot f_0 / Q_{target})] / \alpha$$

Hardware Limits (Fail-Safe):

Even with dynamic prediction, always apply the hardware limits (`pids/2/limitlower` and `limitupper`) to ± 0.5 V. This acts as a hard physical fail-safe to prevent damage if the initial slope estimate slightly overshoots.

Demodulator Settings for HIPIMS DAQ Operation

Application: HIPIMS (High Power Impulse Magnetron Sputtering) with

PLL active
Q-Control active
AGC active
DAQ streaming on Demod 2
Example resonator: ~6 MHz QCM, native Q ≈ 50k
Instrument: Zurich Instruments UHFLI

The goal is to capture fast plasma transients while maintaining stable multi-loop control.

Recommended Demodulator Configuration

Demodulator	Role	Recommended Time Constant	Filter Order	Purpose
Demod 1	PLL Source	0.3 – 1 ms	4	Stable phase tracking without reacting to plasma spikes
Demod 2	DAQ Measurement	.1 – 1 μs	1 (Critical)	High temporal resolution, minimal filter delay
Demod 3	Q-Control Source	20 – 50 μs	4	Fast damping control without noise amplification
Demod 4	AGC Source	50 – 200 ms	4	Slow amplitude stabilization; must not follow pulses

Design Rationale

Demod 1 — PLL Source

The PLL must track real frequency shifts but ignore fast plasma-induced noise.

- Too fast (<100 μs) → PLL reacts to plasma transients → instability.
- Too slow (>5 ms) → PLL may temporarily lose lock during large frequency shifts.

A range of **0.3–1 ms** provides stable tracking for most HIPIMS regimes.

Demod 2 — Measurement (DAQ)

This demodulator captures transient behavior during the HIPIMS pulse.

- Time constant must be **much smaller than the plasma rise time**.
- Filter order must be **1** to minimize group delay distortion.
- If plasma rise ≈ 5 μs → use ~2 μs.

If plasma rise $\approx 20\ \mu\text{s}$ $\rightarrow$ use 5–10 μs .

This demodulator should always be the fastest in the system.

Demod 3 — Q-Control

Q-control implements active viscous damping.

Too fast $\rightarrow$ amplifies noise.

Too slow $\rightarrow$ damping becomes ineffective during fast transients.

A range of **20–50 μs** provides stable damping for a 6 MHz QCM.

Demod 4 — AGC

AGC maintains constant oscillation amplitude.

It must not respond to:

Individual HIPIMS pulses

Plasma ignition spikes

Time constant must be **much larger than pulse duration**.

Example:

Pulse duration = 100 μs

Recommended AGC TC $\geq 50\ \text{ms}$

Loop Hierarchy Requirement

For stable multi-loop operation:

$$\tau(\text{Demod 2}) \ll \tau(\text{Demod 3}) \ll \tau(\text{Demod 1}) \ll \tau(\text{Demod 4})$$

Fast $\rightarrow$ Slow ordering prevents loop interaction and instability.

Important Measurement Note

While Q-control is engaged, the measured damping is:

$$\Gamma_{\text{effective}} = \Gamma_{\text{native}} - \alpha \cdot K_p$$

Thus, during HIPIMS operation:

Frequency shifts are physically meaningful.

Measured Q reflects the actively controlled system.

Native damping requires disabling PID 3 and performing a partial ring-down transient.
